

provide both functional and object-oriented capabilities. It is also interoperable with Java, meaning it can easily access Java code and vice versa since it runs on the Java Virtual Machine (JVM). Kotlin supports higher-order functions that can either accept a function as a parameter, return a function, or do both. It also features smart casting, which automatically typecasts immutable values, and inserts the value in its safe cast. Kotlin offers fast compilation time and excellent tooling support, including IntelliJ IDEA, Eclipse, and Android Studio.

For instance, Kotlin's code is much more concise compared to other object-oriented programming languages. It also provides safety from null pointer exceptions by supporting nullability as part of its system. Furthermore, Kotlin has rich sets of methods, including lazy evaluation, operator overloading, and higher-order functions, to name a few. In addition, Kotlin's tool-friendly nature makes it easy to write code in any Java Integrated Development Environment (IDE), and developers can run Kotlin programs from the command line.

Advantages of Kotlin

Kotlin, the statically typed programming language, has several advantages:

- **Ease of learning** - As Kotlin's basics are almost similar to Java, anyone who has worked with Java can understand Kotlin easily.
- **Multi-platform support** - Kotlin is supported by all Java IDEs, which means you can write your program and execute it on any machine that supports JVM.
- **Safety** - Kotlin provides a safer programming environment compared to Java.
- **Interoperability** - Kotlin allows using Java frameworks and libraries in new Kotlin projects without requiring a complete overhaul of the existing Java project.
- **Open source** - The Kotlin programming language, including the compiler, libraries, and all the tooling, is completely free and open source, and is available on Github at <https://github.com/JetBrains/kotlin>.

Kotlin is a versatile programming language with a range of applications, including:

